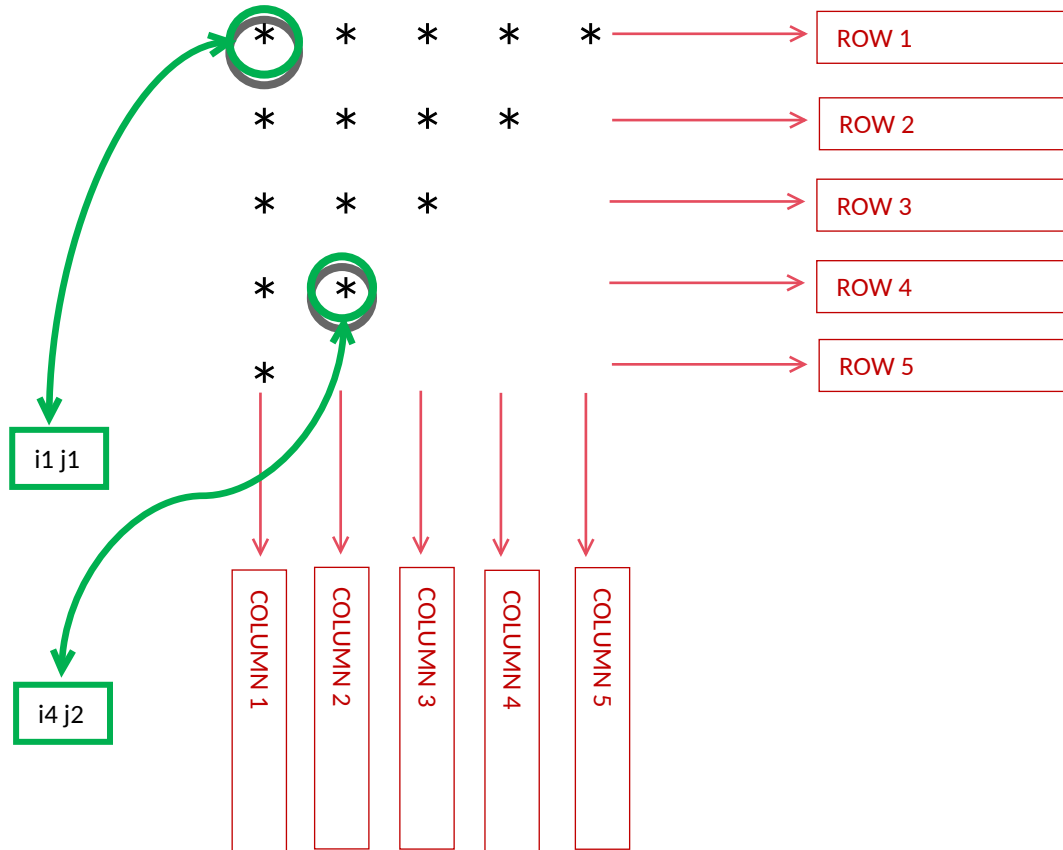


WEEK 01 MODULE
Programming Practice

TOPIC : 1 'Star Pattern'

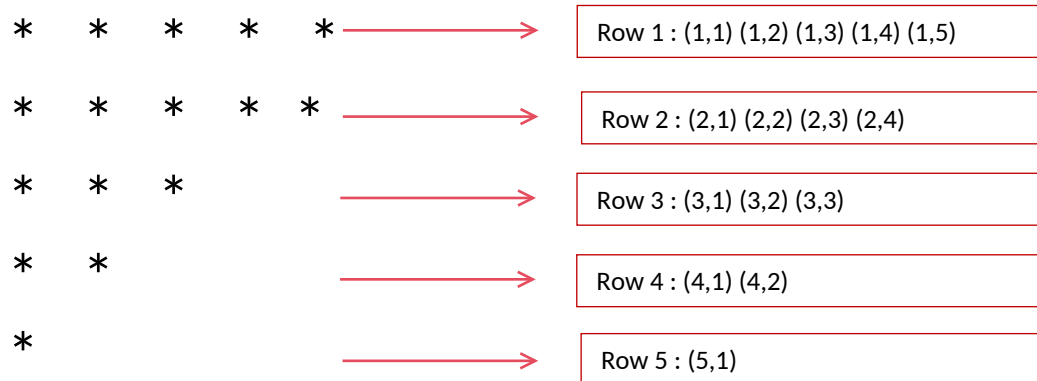
6. HALF DIAMOND(LOWER) $\longleftrightarrow$ INVERTED TRIANGLE



1. Problem Analysis :-

Pattern Structure:

- The output forms an inverted half diamond or half diamond(lower)
- Unlike the Triangle Pattern where the number of stars increases in each row, this pattern decreases the number of stars by one in every subsequent row.
- The pattern starts with n stars in the first row and decreases by one star in each subsequent row until the last row contains one star.
- For n=5



i. Execution Flow:

Step 1: Observe the Grid Concept:

- The pattern can be visualized as a two-dimensional grid (matrix)
- size = $n \times n$.
- Each cell in the grid represents a unique coordinate (i, j), where i denotes the row index and j denotes the column index.
- For $n = 5$:

As the row number increases by 1, the number of stars decreases by 1.

Observation:

- Row 1 → 5 star
-
- Row 2 → 4 stars
-
- Row 3 → 3 stars
-
- Row 4 → 2 stars
-
- Row 5 → 1 stars
-

A clear relationship can be observed: Number of stars = $n - i + 1$

Therefore:

- Row 1 prints n stars
- Row 2 prints n-1 stars
- Row 3 prints n-2 stars

-
- Row n prints 1 star

- The number of valid columns is decreasing.
- Every new row removes one column from the right side.
- Therefore the inner loop cannot remain fixed.
- its limit must shrink with every row.

Step 2: Traverse the grid row by row.

- The outer loop is responsible for generating rows.

Step 3: For every row generated by the outer loop, determine how many stars need to be printed.

- The number of stars depends on the current row number.
- The relationship is:

- Number of Stars = $n - i + 1$

Step 4: Traverse the required number of columns using the inner loop.

- The inner loop is responsible for generating stars in the current row.
- The loop executes from:

- 0 to $(n - i - 1)$

Step 5: At each iteration of the inner loop, print a star (*).

Step 6: After all stars of the current row have been printed, move the cursor to the next line.

Step 7: Repeat the same process until all n rows have been completed.

Half Diamond Lower follows the same logic as the Inverted Triangle Pattern.

2. Program :-

```
package week1_module;
import java.util.Scanner;
public class StarHlafDiamondLower {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of rows: ");
        int n = sc.nextInt();

        for(int i = 0; i < n; i++) {

            for(int j = 0; j < n - i; j++) {
                System.out.print("* ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

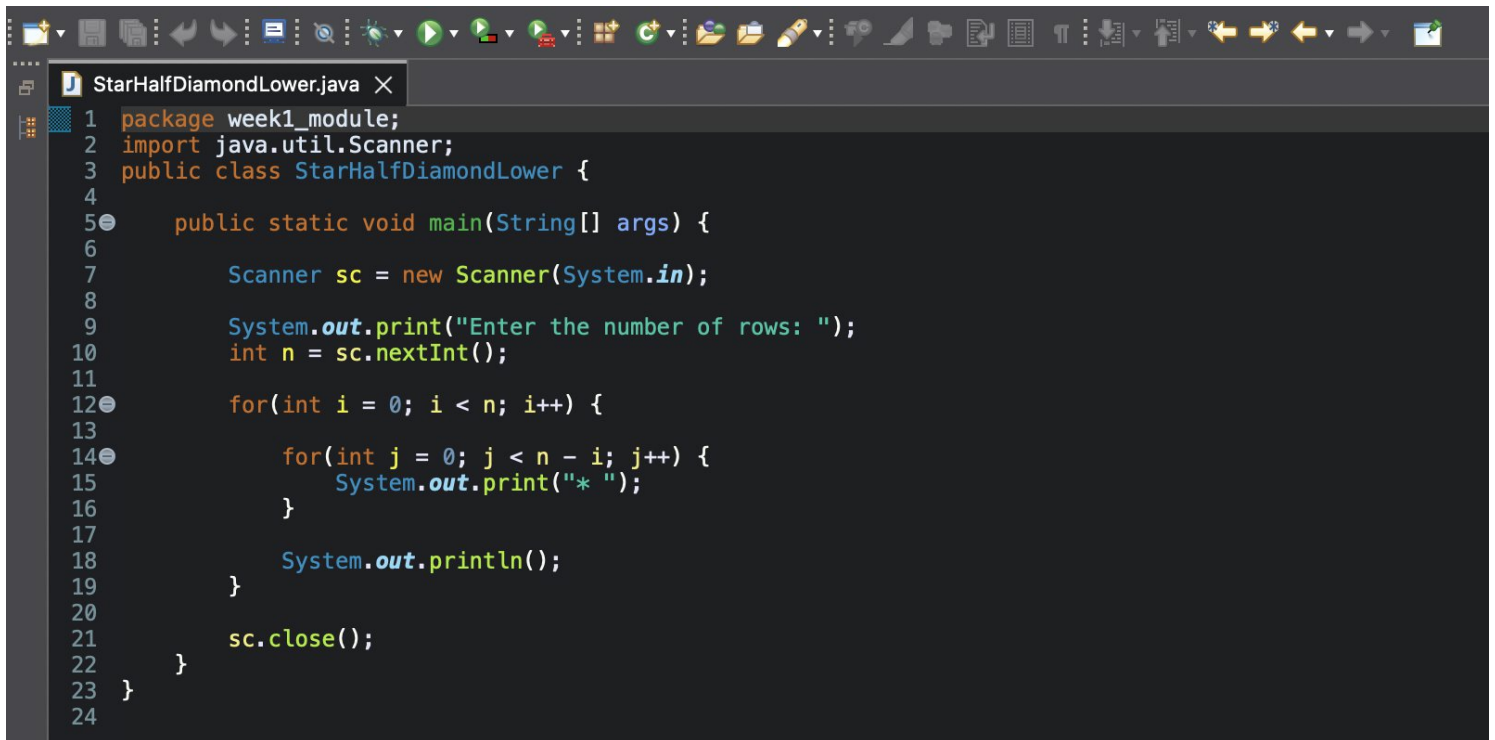
3. Program Output :-

Enter the number of rows: 5

```
* * * * *
* * * *
* * *
* *
*
```

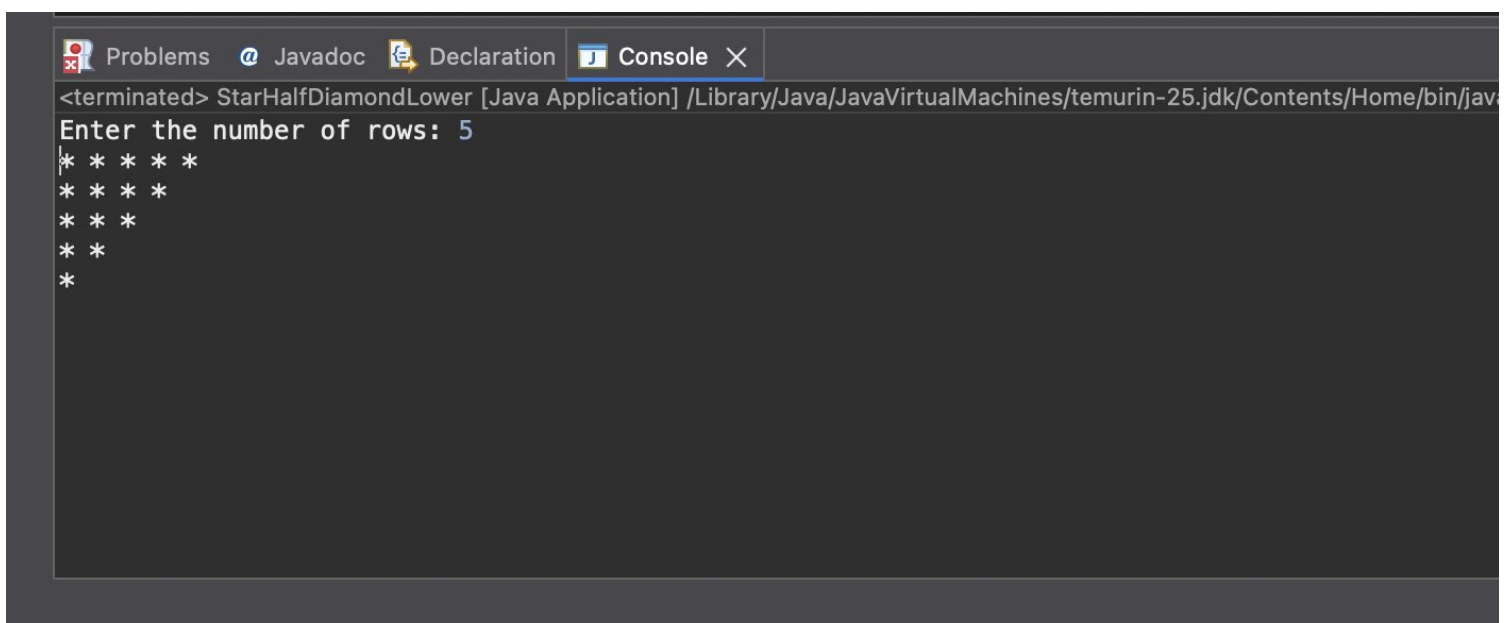
4. ScreenShot :-

i. Program :-



```
1 package week1_module;
2 import java.util.Scanner;
3 public class StarHalfDiamondLower {
4
5     public static void main(String[] args) {
6
7         Scanner sc = new Scanner(System.in);
8
9         System.out.print("Enter the number of rows: ");
10        int n = sc.nextInt();
11
12        for(int i = 0; i < n; i++) {
13
14            for(int j = 0; j < n - i; j++) {
15                System.out.print("* ");
16            }
17
18            System.out.println();
19        }
20
21        sc.close();
22    }
23 }
24
```

ii. Program's Output :-



```
<terminated> StarHalfDiamondLower [Java Application] /Library/Java/JavaVirtualMachines/temurin-25.jdk/Contents/Home/bin/jav
Enter the number of rows: 5
* * * * *
* * * *
* * *
* *
*
```